

ISHURI

AI-Powered School Management System

System Requirements & Design Documentation

Case Study Institution:

Les Hirondelles de Don Bosco – Ndera

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1. Introduction

Ishuri is an AI-powered school management system developed as a case study for Les Hirondelles de Don Bosco, a high school located in Ndera, Rwanda. The name 'Ishuri' is the Kinyarwanda word for 'school', reflecting the system's cultural grounding and its purpose.

The system is designed to support both the school administration and the ICT department in digitizing and streamlining academic and administrative processes that are currently managed manually. Ishuri replaces paper-based workflows — such as physical timetables, handwritten reports, and in-person fee confirmations — with an integrated digital platform accessible to four types of users: administrators, teachers, parents, and students.

The AI component of Ishuri is focused specifically on performance analytics: it analyzes student grades and attendance data to classify students into one of three performance zones (satisfactory, near danger, or danger), and supports administrative decision-making through data-driven reports. The AI does not govern communication between users, user authentication, or financial transactions.

2. System Overview

2.1 System Identity

System Name	Ishuri – AI-Powered School Management System
Case Study Institution	Les Hirondelles de Don Bosco – Ndera
Target Departments	School Administration & ICT Department
User Roles	Administrator, Teacher, Parent, Student
AI Scope	Academic performance analytics, disciplinary analytics, risk prediction
Out of Scope	Online fee payment processing, student assignment submission
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2.2 Objectives

Ishuri is built to achieve the following objectives:

- Eliminate paper-based academic and administrative processes at Les Hirondelles de Don Bosco.
- Provide students and parents with real-time visibility into academic performance and disciplinary standing.
- Enable teachers to digitally upload grades, record attendance, share assignments, and communicate with parents.
- Empower the administration with full oversight of academic data, user management, fee status, and system reports.
- Apply AI-driven analytics to flag students at risk and enable proactive academic intervention.
- Replace physical timetables with auto-generated digital schedules for teachers.

2.3 User Roles Summary

Role	Description	Key Privileges
Administrator	Full system control: manages users, approves data, and generates reports.	Highest — unrestricted access
Teacher	Manages grades, attendance, assignments; communicates with parents.	Class & subject-scoped
Parent	Views child performance, communicates with teachers, monitors fees.	Linked to their children only
Student	Views own performance, assignments, fees; downloads digital ID.	Personal data only

3. Authentication & Login

3.1 User Creation Policy

There is no self-registration in Ishuri. All user accounts are created exclusively by the system administrator. When an account is created, the administrator assigns the user their credentials (username and password) and their role. This design eliminates the need for role-validation workflows and prevents unauthorized access.

Note: The only exception to admin-created accounts is the parent account, which is auto-generated by the system when a student completes their first-login form. See Section 5.3 for details.

3.2 Login Flow

When a user navigates to the Ishuri login page, the following process applies:

- The user enters their credentials (username and password).
- The user selects their role from a dropdown: Student, Teacher, or Parent.
- The system validates both the credentials and the selected role simultaneously.
- If credentials are correct but the role is wrong, access is denied and an error message is shown.
- If both credentials and role are correct, the user is redirected to their respective dashboard.
- The administrator logs in without role selection, as there is only one admin role.

Security Note: Role validation at login prevents a teacher from accessing a student dashboard (or vice versa) even if credentials are somehow compromised.

4. AI Performance Analytics

4.1 Overview

The AI analytics engine in Ishuri continuously evaluates two dimensions of student performance: academic performance (based on grades uploaded by teachers) and disciplinary performance (based on attendance recorded by teachers). Each dimension is independently classified into one of three color-coded zones.

4.2 Color-Coded Performance Zones

Indicator	Status	Meaning
□ Green	Satisfactory	Student is performing well with no concerns.
□ Yellow	Near Danger Zone	Student is approaching risk; early intervention recommended.
□ Red	Danger Zone	Student is in critical academic or disciplinary risk.

These indicators are visible on the dashboards of students, parents, and within admin reports. The AI model determines zone classification by analyzing grade averages against predefined thresholds and computing attendance rates over a rolling period.

4.3 Analytics Scope

- Individual student academic performance (per subject and overall).
- Individual student disciplinary performance (attendance-based).
- Class-level performance aggregates (visible to admin only).
- Risk zone predictions to support early academic intervention.

5. Student Module

5.1 Dashboard Overview

The student dashboard is the primary interface for a student to monitor their academic and disciplinary standing. It is personalized, data-driven, and updated in real time as teachers upload new information.

5.2 Dashboard Features

Feature	Description	Access
Academic Performance Chart	Graphical representation of the student's grades per subject, color-coded (green/yellow/red) based on AI zone classification.	Student
Disciplinary Performance Chart	Graphical representation of the student's attendance rate, color-coded using the same green/yellow/red system.	Student
Assignment Notifications	Displays whether a teacher has uploaded a new assignment for the student's class. The student can view and download the PDF.	Student
Fee Payment Status	Shows the current status of the student's school fees (Paid / Partially Paid / Unpaid). If fees are fully paid and approved, the View Academic Report option becomes available.	Student
Academic Report	Full academic report visible and downloadable (by admin only) once the admin approves fee payment. Students can view but not download.	Student (view only)
Digital ID Card	Auto-generated digital identity card for the student, produced by the system upon account creation.	Student
Password Change	Student can update their login password from the dashboard settings.	Student
Light / Dark Mode Toggle	Student can switch between light and dark interface modes for visual comfort.	Student

5.3 First Login – Parent Account Creation

When a student logs into Ishuri for the first time, the system detects that no parent account is linked to their profile and prompts them to complete a mandatory registration form. This form collects the following information about their parent or legal guardian:

- Full name (both names)
- Physical address
- Email address
- Phone number

Upon submission, the system performs the following actions:

- It checks whether a parent account already exists with the submitted email address or phone number.
- If no existing account is found: a new parent account is created using the submitted details. Credentials are generated and delivered to the parent.
- If an existing account is found (e.g., an older sibling already registered the same parent): the student is simply linked to the existing parent account. No duplicate account is created.

This mechanism ensures that a parent with multiple children at the school has a single unified account from which they can monitor all their children.

6. Parent Module

6.1 Dashboard Overview

The parent dashboard provides guardians with a transparent and real-time view of their children's academic and disciplinary performance. Parents do not interact with administrative processes; their access is entirely read-oriented, with the exception of the live communication channel with teachers.

6.2 Multi-Child Navigation

If a parent has more than one child enrolled at Les Hirondelles de Don Bosco, the dashboard displays all linked children. The parent can toggle between each child's profile using a navigation control. Each child's data is shown independently.

6.3 Dashboard Features

Feature	Description	Access
Child Performance Dashboard	Upon selecting a child's profile, the parent sees the same academic and disciplinary performance charts (green/yellow/red) as the student.	Parent

Academic Report Access	The parent can view the child's full academic report, but only if the school fees have been fully paid and approved by the admin.	Parent (view only, fees-gated)
School Fee Status	Displays the current fee payment status for each linked child (Paid / Partially Paid / Unpaid). Payment itself is not done within the system.	Parent
Teacher Live Chat	Real-time messaging channel enabling the parent to communicate with any teacher who teaches a subject in their child's class.	Parent

6.4 Teacher Communication Scope

The parent's live chat feature is scoped to their child's teaching staff. If a student is enrolled in 8 subjects, the parent has access to exactly those 8 teachers' chat channels. The parent cannot initiate contact with teachers from other classes or subjects unrelated to their child.

Note: Payment of school fees is acknowledged and tracked within Ishuri, but the actual payment transaction takes place outside the system (e.g., bank transfer, cash payment at the school). The admin manually updates the fee status in the system after confirming payment.

7. Teacher Module

7.1 Overview

Each teacher in Ishuri is assigned to one or more classes and subjects. This assignment defines the scope of their entire experience in the system: which students they manage, which grades they upload, which attendance they record, and which parents they communicate with. At Les Hirondelles de Don Bosco, a subject-class pairing unambiguously identifies the responsible teacher — for instance, a student in Grade 10 studying Chemistry has one teacher for that subject.

7.2 Dashboard Features

Feature	Description	Access
Auto-Generated Timetable	The system generates a daily timetable for each teacher based on their assigned courses and classes. This replaces paper timetables. The timetable shows the teacher's class schedule for the current day.	Teacher

Grade Upload	The teacher uploads student grades in the form of an Excel file. Uploaded grades are submitted to the admin for approval before appearing on student reports.	Teacher
Attendance Recording	The teacher records student attendance for their class. This data feeds the student's disciplinary performance chart.	Teacher
Assignment Upload	The teacher can upload a PDF document as an assignment for their students. Students can view and download the assignment; submission is done physically in class.	Teacher
Parent Live Chat	Real-time messaging with parents of students in the teacher's class(es).	Teacher

7.3 Grade Upload Workflow

The following describes the end-to-end flow for grade submission:

- The teacher prepares grades in an Excel (.xlsx) file formatted according to the system's required template.
- The teacher uploads the file through their dashboard.
- The system registers the upload and notifies the administrator of a pending grade review.
- The administrator reviews the uploaded grades.
- If approved: grades are published to student and parent dashboards and integrated into academic reports.
- If not approved or if there is an issue: the administrator can flag the submission and notify the teacher to revise and re-upload.

Grades do not appear on any student-facing view until explicitly approved by the administrator. This ensures data quality and gives the administration oversight over academic records.

7.4 Attendance as Dual-Purpose Record

When a teacher records student attendance in Ishuri, this action serves two simultaneous purposes:

- **Student Discipline Tracking:** The attendance data is used by the AI engine to compute the student's disciplinary performance score and assign a green, yellow, or red zone classification.
- **Teacher Attendance Verification:** The act of uploading attendance is itself treated as confirmation that the teacher was present and conducted the class. The admin can use this to generate teacher attendance reports and identify unreported absences.

This dual-purpose design means that a teacher who does not upload attendance for a class is flagged both as potentially absent and as failing to provide disciplinary data for their students.

8. Administrator Module

8.1 Overview

The administrator holds the highest level of privilege in Ishuri. The admin is responsible for user management, academic data governance, fee status control, and system reporting. The admin account is not created through any user-facing form — it is provisioned directly at the time of system setup.

8.2 User Management

The administrator is the sole authority for creating, editing, and removing user accounts in the system. This covers all four user types: admins, teachers, parents (when not auto-created via student first login), and students. When creating an account, the admin assigns:

- Username and initial password
- User role (student, teacher, parent, or admin)
- For teachers: assigned class(es) and subject(s)
- For students: class enrollment

This design removes the self-signup flow entirely, ensuring that only verified, enrolled individuals gain access to the system.

8.3 Grade Approval

As described in Section 7.3, the admin reviews all teacher-uploaded grade files before they are published. This gives the administration oversight over academic records and prevents unauthorized or erroneous data from appearing on student reports.

8.4 Fee Status Management

The admin controls the school fee payment status for every student. After confirming that a payment has been received (through the school's external payment channel), the admin updates the fee status in Ishuri to one of:

- Unpaid – Student has not made any payment.
- Partially Paid – Student has made an incomplete payment.
- Fully Paid – Payment confirmed; academic report access is unlocked.

Only when the status is set to Fully Paid will the View Academic Report option become available on the student and parent dashboards. The admin uses this mechanism to ensure that outstanding fees are settled before academic records are released, as releasing reports to students who still owe fees could reduce the school's ability to collect payment.

8.5 Report Downloads

Academic reports can only be downloaded by the administrator. Students and parents may view reports on-screen (subject to fee status), but they do not have download or export access. This restriction prevents students or parents from obtaining a portable copy of their report while fees remain outstanding — which could otherwise be used to enroll elsewhere without settling debts.

8.6 System Reports

The admin has access to the following generated reports:

Report Type	Description	Accessible By
Class Average per Subject	Mean grade per subject in each class	Admin
Teacher Attendance Rate	Tracks frequency of class attendance uploads by teachers	Admin
Students in Risk Zone	Lists students flagged red in academic or disciplinary metrics per class	Admin
Fee Collection Status	Summary of paid, partial, and unpaid school fees per student/class	Admin
Login Activity Report	Tracks user logins by role, frequency, and timestamps	Admin
Student Enrollment List	Complete roster of enrolled students with class assignments	Admin
General School Performance Report	Aggregate academic performance across all classes and subjects	Admin

9. Digital Student ID Card

Upon account creation, each student is automatically issued a digital ID card by the system. The card is accessible from the student's dashboard and contains identifying information linked to their enrollment record.

The digital ID card serves as a modern replacement for physical student cards and can be displayed digitally when required for school identification purposes.

10. Live Communication Channel

10.1 Overview

Ishuri includes a real-time live chat feature to facilitate communication between parents and teachers. This is not an AI chatbot — it is a direct person-to-person messaging channel, similar in concept to a messaging application.

10.2 Communication Rules

- A parent can initiate a conversation with any teacher who teaches a subject to their child.
- A teacher can receive messages from any parent whose child is in one of the teacher's assigned classes.
- Chat is scoped by class and subject — cross-class or unrelated teacher access is not permitted.
- All chat history is persistent within a session and logged in the system.

11. Auto-Generated Teacher Timetable

The system automatically generates a daily class schedule for each teacher based on their subject and class assignments. This timetable is displayed on the teacher's dashboard and reflects the classes they are scheduled to teach on the current day.

This feature replaces the physical paper timetables that teachers at Les Hirondelles de Don Bosco previously carried by hand, reducing administrative overhead and ensuring that schedule updates are immediately reflected for all affected teachers.

12. Key Data Flows

12.1 Grade Data Flow

- Teacher uploads Excel grade file
 - System registers pending review
 - Admin is notified
- Admin reviews and approves or flags
 - If approved: grades are published to student & parent dashboards and integrated into reports

- If flagged: teacher is notified to revise and re-upload

12.2 Attendance Data Flow

- Teacher records attendance for their class
- Attendance is simultaneously used to:
 - Compute student disciplinary score (AI analytics)
 - Log teacher's own class presence for admin reports

12.3 Fee & Report Access Flow

- Admin receives confirmation of payment (external)
- Admin updates fee status in Ishuri to 'Fully Paid'
- System unlocks 'View Academic Report' on student and parent dashboards
- Admin retains exclusive download privilege

12.4 Parent Account Creation Flow

- Student logs in for the first time
- System detects no linked parent account
- Student completes parent/guardian registration form
- System checks for existing account by email or phone
 - If match found: student is linked to existing parent account
 - If no match: new parent account is created and linked

13. Scope & Known Limitations

13.1 In Scope

- Role-based user management (admin-created accounts)
- Student academic and disciplinary performance dashboards
- AI-powered performance zone classification (green/yellow/red)
- Teacher grade upload with admin approval workflow
- Attendance recording with dual-purpose logging
- PDF assignment distribution (view & download, no submission)
- Fee status management and report access gating
- Auto-generated teacher daily timetable
- Real-time parent-teacher live chat
- Digital student ID card generation

- Admin system performance reports

13.2 Out of Scope

- Online fee payment processing — payments are handled externally; the system only reflects status after admin confirmation.
- Assignment submission — students submit assignments physically; the system only distributes them.
- Parent or student self-registration — all accounts are admin-managed (except parent auto-creation via first-login form).
- AI-driven chat or chatbot responses — the communication channel is strictly human-to-human.

14. Glossary

Term	Definition	Notes
Ishuri	Kinyarwanda for 'school'. The name of this system.	—
Performance Zone	AI classification of a student's status: green (satisfactory), yellow (near danger), red (danger).	—
Fees Gating	The restriction of academic report access until school fees are fully paid and approved by admin.	—
Grade Upload	The process by which a teacher submits an Excel file of student grades for admin review.	—
Attendance Upload	The process of recording student attendance; also logs teacher class presence.	—
First Login Form	A mandatory form completed by students on their first login, used to create the linked parent account.	—
Digital ID Card	An auto-generated digital identity card issued to each student upon account creation.	—
Live Chat	Real-time, human-to-human messaging between parents and teachers within the system.	—
Admin Reports	System-generated data summaries accessible only to the administrator.	—

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